

M4b: discharge audit and corrected proof program for continuum LPPL in $P(\varphi)_2$

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Abstract

This note audits, one by one, the four inputs left open by the conditional LPPL theorem in `lppl.tex`. Two conclusions are unconditional within the standard spatial-cutoff $P(\varphi)_2$ construction. First, the actual cutoff dynamics has propagation speed at most one; finite propagation is therefore a theorem, not a hypothesis. Second, every cutoff ground-state vector belongs to the operator domain of every compactly smeared Wick polynomial. This removes the vector-domain part of the differentiability input and the domain clause from the moment input.

The other conclusions are negative but precise. A common quadratic-form domain on the entire closed interpolation interval is not a legitimate generic assumption when an interaction is turned on from zero. It is replaced below by local form differentiability on $0 < s < 1$ and endpoint norm-resolvent continuity. The proposed modern log-Sobolev input has the unweighted stochastic-quantisation Dirichlet form, not the covariance-weighted form used in `h6-attack.tex`; it therefore proves neither the physical Hamiltonian gap nor the required local second moment. Moreover, the induction in `h6-attack.tex` contains two false operator estimates and is withdrawn. A corrected conditional induction is given, exposing the additional one-point bound that would actually be needed.

Thus the continuum LPPL argument itself is complete, exact propagation is complete, and the ground-state domain question is complete. The affine path regularity is also proved below from the Feynman–Kac–Nelson formula once the same uniform gap already assumed by LPPL is available. Two sharply stated analytic obligations remain: a uniform physical gap in a specified noncritical regime and a uniform local second moment. Neither is asserted without proof.

1 Setting and verdict

Let

$$K(g) = H_0 + V(g), \quad V(h) = \int_{\mathbb{R}} h(x) : P(\varphi_0(x)) : dx, \quad H(g) = K(g) - E(g),$$

where $E(g) = \inf \text{spec } K(g)$ and

$$\mathcal{G} = \{g \in C_c^\infty(\mathbb{R}) : 0 \leq g \leq 1, \quad g = 1 \text{ on a nonempty bounded interval}\}.$$

The ground state of $H(g)$ is denoted by Ω_g and is normalized. The time-zero local algebra over a bounded open interval O is $\mathfrak{A}(O)$, and

$$\beta_t^g(A) = e^{itH(g)} A e^{-itH(g)}.$$

All assertions below use the standard Q -space construction of the spatially cutoff $P(\varphi)_2$ Hamiltonian: $V(h)$ is the self-adjoint multiplication operator associated with the Wick polynomial, and the usual $L^1 \cap L^2$ extension of the cutoff is allowed when a characteristic function is used in a localization argument.

Input in <code>lpp1.tex</code>	Verdict	Exact remainder
finite propagation (L)	proved, Proposition 2.1	none
$\Omega_g \in D(V(h))$	proved, Proposition 3.1	none
path differentiability (D)	follows from (G)	no independent input, §4
uniform physical gap (G)	open	(G_{exact}), §5
uniform local moment (M)	open	(M_2), §6

The word “open” in the table is mathematical, not procedural: the sources audited here do not contain the required gap or moment theorem, and the arguments previously proposed in this folder do not prove either one. The conditional LPPL theorem remains valid after finite propagation and the vector-domain condition are removed from its hypothesis list; Theorem 4.6 also removes path differentiability as an independent hypothesis.

2 Exact finite propagation for the cutoff dynamics

We spell out the localization argument because the LPPL cell expansion uses the cutoff dynamics far outside the plateau $\{g = 1\}$.

Proposition 2.1 (cutoff propagation). *For every $g \in \mathcal{G}$, every bounded open interval $O \subset \mathbb{R}$, and every $t \in \mathbb{R}$,*

$$\beta_t^g(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|}), \quad O_r = \{x : \text{dist}(x, O) < r\}.$$

The speed bound is one and is independent of g . No assumption that $g = 1$ on $O_{|t|}$ is needed for this inclusion.

Proof. The scalar $E(g)I$ cancels from the adjoint action, so it suffices to use $K(g) = H_0 + V(g)$. Write

$$\beta_t^0 = \text{Ad } e^{itH_0}, \quad \beta_t^{I,g} = \text{Ad } e^{itV(g)}.$$

First, $\beta_t^{I,g}$ has propagation speed zero. It is enough initially to take an observable A generated by Weyl operators with test functions supported in a compact set $B \Subset O$. Choose $\chi \in C_c^\infty(O)$ with $\chi = 1$ on a neighbourhood of B and split

$$V(g) = V(\chi g) + V((1 - \chi)g).$$

The two summands strongly commute because they are multiplication operators in Q -space. Affiliation gives

$$e^{itV(\chi g)} \in \mathfrak{A}(O), \quad e^{itV((1-\chi)g)} \in \mathfrak{A}(B)'.$$

Consequently

$$\beta_t^{I,g}(A) = e^{itV(\chi g)} A e^{-itV(\chi g)} \in \mathfrak{A}(O).$$

Strong density of the compactly supported Weyl-generated subalgebra and strong closedness of $\mathfrak{A}(O)$ extend this conclusion to every $A \in \mathfrak{A}(O)$.

Second, the free Klein–Gordon equation has propagation speed one, hence

$$\beta_t^0(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|}).$$

For $A \in \mathfrak{A}(O)$, the n th Trotter approximant

$$A_n(t) = (\beta_{t/n}^0 \circ \beta_{t/n}^{I,g})^n(A)$$

therefore belongs to $\mathfrak{A}(O_{|t|})$: each interaction step leaves the current localization region unchanged, and each free step enlarges it by $|t|/n$. The self-adjointness theorem for $H_0 + V(g)$ and the Trotter product formula give

$$e^{itK(g)} = \text{s-}\lim_{n \rightarrow \infty} (e^{itH_0/n} e^{itV(g)/n})^n.$$

The same holds for the adjoints. Hence $A_n(t) \rightarrow \beta_t^g(A)$ strongly. Since $\mathfrak{A}(O_{|t|})$ is a von Neumann algebra and therefore strongly closed, $\beta_t^g(A) \in \mathfrak{A}(O_{|t|})$. $\square$

Remark 2.2. This is the argument of Glimm–Jaffe’s finite-propagation theorem [2, Ch. 4, Prop. 4.1.1 and Thm. 4.1.2]; their later formulation [3, Thm. 8.1] also proves cutoff independence when $g = 1$ on the relevant light cone. Cutoff independence is an additional conclusion. The propagation inclusion itself follows from “free speed one + interaction speed zero” for every spatial cutoff g .

3 The ground state is in every local Wick-polynomial domain

Proposition 3.1 (domain smoothing). *Let $g \in \mathcal{G}$ and let $h \in C_c^\infty(\mathbb{R})$ be real. Then*

$$\Omega_g \in D(V(h)).$$

In particular, for an affine path $g_s = (1-s)g + sg'$ and $v = g' - g$, $\Omega_{g_s} \in D(V(v))$ for every $0 \leq s \leq 1$.

Proof. The self-adjointness and smoothing theorem for cutoff $P(\varphi)_2$ Hamiltonians states that, if $K = H_0 + V(g)$ and V' is any compactly smeared Wick polynomial in the standard polynomial $L^2(Q)$ class, then for every $t > 0$,

$$\text{Ran } e^{-tK} \subseteq D(H_0) \cap D(V').$$

See [3, Ch. 7, Thm. 7.3 and Lem. 7.4]. Take $V' = V(h)$. Since $K(g)\Omega_g = E(g)\Omega_g$,

$$\Omega_g = e^{tE(g)} e^{-tK(g)} \Omega_g.$$

The right-hand side belongs to $D(V(h))$ by the displayed range inclusion. This proves the claim. Taking $h = v$ gives the path statement. $\square$

Thus the domain clause in hypothesis (M) is redundant. The unresolved content of (M) is solely the numerical estimate $\sup_g \|V(h)\Omega_g\| < \infty$ for unit-cell h .

4 The correct differentiability input

4.1 Why a global common form domain is not available automatically

The original hypothesis (D) in `lpp1.tex` imposed one common form domain for all $0 \leq s \leq 1$. Turning on a positive highest-degree interaction can change the form domain at an endpoint. This is already visible in one-dimensional quantum mechanics.

Proposition 4.1 (explicit endpoint obstruction). *On $L^2(\mathbb{R})$ let*

$$K_s = -\frac{d^2}{dx^2} + x^2 + sx^4, \quad 0 \leq s \leq 1,$$

with the Friedrichs realization. Its form domains satisfy

$$D(q_0) = \{\psi \in H^1(\mathbb{R}) : x\psi \in L^2\},$$

whereas, for every $s > 0$,

$$D(q_s) = \{\psi \in H^1(\mathbb{R}) : x\psi \in L^2, x^2\psi \in L^2\}.$$

In particular the domains are not common on $[0, 1]$.

Proof. The two identities follow directly from the nonnegative closed forms

$$q_s[\psi] = \|\psi'\|_2^2 + \|x\psi\|_2^2 + s\|x^2\psi\|_2^2.$$

To see that the inclusion is strict, put $\psi(x) = (1 + x^2)^{-1}$. Then $\psi, \psi', x\psi \in L^2(\mathbb{R})$: at infinity their absolute values are respectively $O(|x|^{-2})$, $O(|x|^{-3})$, and $O(|x|^{-1})$. But

$$|x^2\psi(x)|^2 = \frac{x^4}{(1 + x^2)^2} \longrightarrow 1,$$

so $x^2\psi \notin L^2(\mathbb{R})$. Hence $\psi \in D(q_0) \setminus D(q_s)$ for $s > 0$. $\square$

This does not show that the $P(\varphi)_2$ ground state is discontinuous. It shows only that a proof may not assume a global common domain without a separate theorem.

4.2 A minimal replacement

Let $g, g' \in \mathcal{G}$ have a common plateau, $g_s = (1 - s)g + sg'$, $v = g' - g$, and write $K_s = K(g_s)$, $E_s = E(g_s)$, $H_s = K_s - E_s$, and $P_s = |\Omega_s\rangle\langle\Omega_s|$.

Hypothesis 4.2 (minimal path regularity, $D_{\min}$). *The following two statements hold.*

(D1) *For every $s_0 \in (0, 1)$ there are an open interval $I \ni s_0$ and a Hilbert form domain $\mathcal{Q}_I \hookrightarrow \mathcal{H}_0$, common to $s \in I$, such that the closed forms of K_s obey*

$$k_s = k_{s_0} + (s - s_0)v_I,$$

where v_I is a bounded symmetric form on $\mathcal{Q}_I$. On the ground state it is represented by the operator $V(v)$:

$$v_I(\xi, \Omega_s) = \langle \xi, V(v)\Omega_s \rangle \quad (\xi \in \mathcal{Q}_I, s \in I).$$

(D2) *The shifted Hamiltonians converge in the strong resolvent sense at both endpoints:*

$$H_s \xrightarrow[s \downarrow 0]{\text{s.r.}} H_0, \quad H_s \xrightarrow[s \uparrow 1]{\text{s.r.}} H_1.$$

The notation H_0 in (D2) means $H(g)$, not the free Hamiltonian. Proposition 3.1 already supplies the operator-domain assertion in (D1).

Lemma 4.3 (local form regularity gives the derivative). *Assume (D1) and suppose*

$$\text{spec } H_s \subseteq \{0\} \cup [\gamma, \infty) \quad (s \in I)$$

for some $\gamma > 0$. After a phase choice, $s \mapsto \Omega_s$ is C^1 locally on I and

$$\dot{E}_s = \langle \Omega_s, V(v)\Omega_s \rangle, \tag{4.1}$$

$$\dot{\Omega}_s = -H_s^{-1}(1 - P_s)V(v)\Omega_s. \tag{4.2}$$

Proof. Give $\mathcal{Q}_I$ the norm $\|\xi\|_{\mathcal{Q}_I}^2 = k_{s_0}[\xi] + c\|\xi\|^2$ with c so large that it is positive. Let $J : \mathcal{Q}_I \rightarrow \mathcal{Q}_I^*$ be the canonical embedding induced by the Hilbert-space inner product, and let $A_s : \mathcal{Q}_I \rightarrow \mathcal{Q}_I^*$ represent k_s . Then

$$A_s = A_{s_0} + (s - s_0)V_I$$

in $B(\mathcal{Q}_I, \mathcal{Q}_I^*)$. If z is in the resolvent set of K_s , the weak resolvent is $(A_s - zJ)^{-1} : \mathcal{Q}_I^* \rightarrow \mathcal{Q}_I$. The inverse identity gives, whenever both sides exist,

$$(A_{s+h} - zJ)^{-1} - (A_s - zJ)^{-1} = -h(A_{s+h} - zJ)^{-1}V_I(A_s - zJ)^{-1}.$$

Thus the resolvent is C^1 as a map $\mathcal{Q}_I^* \rightarrow \mathcal{Q}_I$, with derivative

$$-(A_s - zJ)^{-1}V_I(A_s - zJ)^{-1}.$$

Composing with the continuous embeddings $\mathcal{H}_0 \hookrightarrow \mathcal{Q}_I^*$ and $\mathcal{Q}_I \hookrightarrow \mathcal{H}_0$ gives norm-resolvent C^1 dependence on s .

Fix s_0 and choose a circle Γ enclosing only the simple eigenvalue E_s for all s in a smaller neighbourhood; the gap and norm-resolvent continuity permit this. The Riesz projection

$$P_s = -\frac{1}{2\pi i} \int_{\Gamma} (K_s - z)^{-1} dz$$

is C^1 in $B(\mathcal{H}_0)$ and, by the preceding mapping statement, as a map $\mathcal{H}_0 \rightarrow \mathcal{Q}_I$. The vector

$$\tilde{\Omega}_s = \frac{P_s \Omega_{s_0}}{\|P_s \Omega_{s_0}\|}$$

is therefore C^1 in $\mathcal{Q}_I$ after shrinking the neighbourhood. Multiplication by a C^1 phase puts it in the parallel gauge $\langle \Omega_s, \dot{\Omega}_s \rangle = 0$.

Differentiate the weak eigenvalue equation

$$k_s(\xi, \Omega_s) = E_s \langle \xi, \Omega_s \rangle, \quad \xi \in \mathcal{Q}_I.$$

Taking $\xi = \Omega_s$ after differentiating and using the weak eigenvalue equation once more with test vector $\dot{\Omega}_s$ gives (4.1). For general ξ one obtains

$$(k_s - E_s \langle \cdot, \cdot \rangle)(\xi, \dot{\Omega}_s) = -\langle \xi, (V(v) - \dot{E}_s) \Omega_s \rangle.$$

The right side is represented by a Hilbert-space vector by Proposition 3.1. The representation theorem for closed forms therefore gives

$$H_s \dot{\Omega}_s = -(1 - P_s)V(v)\Omega_s.$$

The parallel gauge puts $\dot{\Omega}_s$ in $(1 - P_s)\mathcal{H}_0$, where H_s^{-1} has norm at most γ^{-1} . This proves (4.2). $\square$

Lemma 4.4 (endpoint projection continuity). *Assume (D2) and a uniform gap $\text{spec } H_s \subseteq \{0\} \cup [\gamma, \infty)$ for $0 \leq s \leq 1$. Then*

$$\|P_s - P_0\| \longrightarrow 0 \quad (s \downarrow 0), \quad \|P_s - P_1\| \longrightarrow 0 \quad (s \uparrow 1).$$

Consequently $s \mapsto \omega_{g_s}(A)$ is continuous at both endpoints for every bounded A .

Proof. Strong resolvent convergence implies $e^{-tH_s} \rightarrow e^{-tH_0}$ strongly for every fixed $t > 0$. The uniform gap gives

$$\|e^{-tH_s} - P_s\| \leq e^{-\gamma t}, \quad \|e^{-tH_0} - P_0\| \leq e^{-\gamma t}.$$

For every $\xi \in \mathcal{H}_0$,

$$\|(P_s - P_0)\xi\| \leq 2e^{-\gamma t}\|\xi\| + \|(e^{-tH_s} - e^{-tH_0})\xi\|.$$

First let $s \downarrow 0$, then $t \rightarrow \infty$; this proves $P_s \rightarrow P_0$ strongly. Since both are rank-one projections,

$$\|P_s - P_0\|^2 = 1 - \text{Tr}(P_s P_0).$$

Strong convergence applied to a unit vector spanning $\text{Ran } P_0$ makes the right-hand side tend to zero. The proof at $s = 1$ is identical. Finally $\omega_{g_s}(A) = \text{Tr}(P_s A)$ and $|\text{Tr}((P_s - P_0)A)| \leq \|P_s - P_0\|_1 \|A\| = 2\|P_s - P_0\| \|A\|$ for rank at most two. $\square$

Corollary 4.5 (what LPPL actually needs). *In `lppl.tex`, global hypothesis (D) may be replaced by $(D_{\min})$. The derivative formula holds for $0 < s < 1$. If its right-hand side is bounded in absolute value by a constant C_A independent of s , then*

$$|\omega_{g'}(A) - \omega_g(A)| \leq C_A.$$

Proof. Lemmas 4.3 and 4.4 give differentiability on $(0, 1)$ and continuity on $[0, 1]$. For $0 < \varepsilon < 1/2$, the ordinary mean-value estimate on $[\varepsilon, 1 - \varepsilon]$ gives

$$|\omega_{g_{1-\varepsilon}}(A) - \omega_{g_\varepsilon}(A)| \leq (1 - 2\varepsilon)C_A.$$

Let $\varepsilon \downarrow 0$. $\square$

The form criterion above is sufficient, but the actual model admits a cleaner route which bypasses form domains altogether.

Theorem 4.6 (FKN path regularity). *Let $g, g' \in \mathcal{G}$ have a common nonempty plateau, let $g_s = (1 - s)g + sg'$, and put $v = g' - g$. Assume the uniform gap*

$$\text{spec } H(g_s) \subseteq \{0\} \cup [\gamma, \infty) \quad (0 \leq s \leq 1).$$

Then the ground-state projections P_s are norm-continuous on $[0, 1]$ and C^1 on $(0, 1)$. Locally on $(0, 1)$, normalized vectors can be chosen C^1 in Hilbert norm and in the parallel gauge, and they satisfy

$$\begin{aligned} \dot{E}_s &= \langle \Omega_s, V(v) \Omega_s \rangle, \\ \dot{\Omega}_s &= -H_s^{-1}(1 - P_s)V(v)\Omega_s. \end{aligned}$$

Moreover $H_s \rightarrow H_0$ in strong resolvent sense as $s \downarrow 0$ and $H_s \rightarrow H_1$ in strong resolvent sense as $s \uparrow 1$. Thus affine-path differentiability is not an independent hypothesis once the LPPL gap hypothesis is assumed.

Proof. Fix a Euclidean time $T > 0$. On free Euclidean path space let J_0, J_T be the time-slice maps and put

$$\mathcal{U}_s = \int_0^T \int_{\mathbb{R}} g_s(x) : P(\varphi(t, x)) : dx dt, \quad w_s = e^{-\mathcal{U}_s}, \quad \mathcal{U}_v = \mathcal{U}_1 - \mathcal{U}_0.$$

The Feynman–Kac–Nelson formula gives, initially for bounded time-zero functions f, h ,

$$\langle f, e^{-TK(g_s)} h \rangle = \int \overline{J_T f} (J_0 h) w_s d\mu_0. \quad (4.3)$$

Let

$$q_T = \frac{2}{1 - e^{-m_0 T}}, \quad r_T = \frac{q_T}{q_T - 1}.$$

For completeness, the time-slice product estimate follows with no hidden interpolation step. Let ν_0 be the time-zero Gaussian marginal of μ_0 , let $P_T = e^{-TH_0}$ be its Markov semigroup, and set

$$r_T = \frac{2}{1 + e^{-m_0 T}}, \quad p_T = \frac{2}{r_T} = 1 + e^{-m_0 T}, \quad p'_T = 1 + e^{m_0 T}.$$

Then p'_T is conjugate to p_T and $(p'_T - 1)/(p_T - 1) = e^{2m_0 T}$. Nelson hypercontractivity therefore gives $\|P_T F\|_{p'_T} \leq \|F\|_{p_T}$. The Markov property and Hölder's inequality now give, first for bounded f, h ,

$$\begin{aligned} \|(J_T f)(J_0 h)\|_{r_T}^{r_T} &= \int |h|^{r_T} P_T(|f|^{r_T}) d\nu_0 \\ &\leq \| |h|^{r_T} \|_{p_T} \|P_T(|f|^{r_T})\|_{p'_T} \\ &\leq \| |h|^{r_T} \|_{p_T} \| |f|^{r_T} \|_{p_T} = \|h\|_2^{r_T} \|f\|_2^{r_T}. \end{aligned}$$

Taking the r_T th root proves

$$\|(J_T f)(J_0 h)\|_{r_T} \leq \|f\|_2 \|h\|_2. \quad (4.4)$$

This is the estimate used in the transfer-matrix construction [6, Thms. V.10 and V.12].

Every compactly smeared Wick polynomial belongs to $L^p(\mu_0)$ for every finite p ; hence $\mathcal{U}_v \in \bigcap_{p < \infty} L^p(\mu_0)$. Since $g_0, g_1 \geq 0$ and P is semibounded, the standard exponential estimate [6, Thm. V.7] gives $e^{-a\mathcal{U}_0}, e^{-a\mathcal{U}_1} \in L^1(\mu_0)$ for every $a > 0$. Linearity $\mathcal{U}_s = (1 - s)\mathcal{U}_0 + s\mathcal{U}_1$ and Hölder's inequality therefore imply

$$\|w_s\|_a^a \leq \left(\int e^{-a\mathcal{U}_0} d\mu_0 \right)^{1-s} \left(\int e^{-a\mathcal{U}_1} d\mu_0 \right)^s \quad (4.5)$$

for every finite a . In particular the right side is bounded uniformly in $0 \leq s \leq 1$.

Because $\mathcal{U}_s - \mathcal{U}_r = (s - r)\mathcal{U}_v$ in every L^p , $w_s \rightarrow w_r$ in probability as $s \rightarrow r$. Applying (4.5) with an exponent strictly larger than a gives uniform integrability of $|w_s|^a$; Vitali's theorem yields

$$\|w_s - w_r\|_a \rightarrow 0 \quad (s \rightarrow r) \quad (4.6)$$

for every finite a .

Fix $q = q_T$ and choose finite p, a with $q^{-1} = p^{-1} + a^{-1}$. Pointwise in path space,

$$\frac{w_{s+h} - w_s}{h} = -\mathcal{U}_v \int_0^1 w_{s+\theta h} d\theta.$$

Using Hölder's inequality and (4.6),

$$\left\| \frac{w_{s+h} - w_s}{h} + \mathcal{U}_v w_s \right\|_q \leq \|\mathcal{U}_v\|_p \sup_{0 \leq \theta \leq 1} \|w_{s+\theta h} - w_s\|_a \rightarrow 0.$$

Thus $s \mapsto w_s$ is C^1 in L^{q_T} on $(0, 1)$, with one-sided continuity at the endpoints, and $\dot{w}_s = -\mathcal{U}_v w_s$. Indeed, continuity of the derivative follows from

$$\|\mathcal{U}_v(w_s - w_r)\|_{q_T} \leq \|\mathcal{U}_v\|_p \|w_s - w_r\|_a \rightarrow 0.$$

Equations (4.3)–(4.4) now give

$$\|e^{-TK(g_s)} - e^{-TK(g_r)}\| \leq \|w_s - w_r\|_{q_T}$$

and the same estimate for difference quotients. Therefore

$$s \mapsto e^{-TK(g_s)}$$

is norm-continuous on $[0, 1]$ and C^1 on $(0, 1)$.

Set $L_s = e^{-TK(g_s)}$. Its largest spectral value is $\lambda_s = \|L_s\| = e^{-TE_s}$, it is simple, and the gap assumption puts the remainder of its spectrum in $[0, e^{-T\gamma\lambda_s}]$. Norm continuity of L_s implies continuity of λ_s , hence of E_s . A local Riesz contour around λ_s then shows that P_s is C^1 on $(0, 1)$ and norm-continuous at both endpoints. A local vector is obtained from $P_s\Omega_{s_0}/\|P_s\Omega_{s_0}\|$; a C^1 phase change imposes $\langle\Omega_s, \dot{\Omega}_s\rangle = 0$. Since $\lambda_s = \langle\Omega_s, L_s\Omega_s\rangle$, the eigenvalue λ_s and therefore $E_s = -T^{-1}\log \lambda_s$ are C^1 on $(0, 1)$.

It remains to identify the derivative. The cutoff Hamiltonians have the common operator core $\mathcal{C} = C^\infty(H_0)$ and, on this core,

$$K(g_s)\xi = K(g_{s_0})\xi + (s - s_0)V(v)\xi \quad (\xi \in \mathcal{C}),$$

see [6, Thm. V.12(a)]. For $\xi \in \mathcal{C}$, differentiate

$$\langle K(g_s)\xi, \Omega_s \rangle = E_s \langle \xi, \Omega_s \rangle.$$

Proposition 3.1 permits $\langle V(v)\xi, \Omega_s \rangle = \langle \xi, V(v)\Omega_s \rangle$, so

$$\langle (K(g_s) - E_s)\xi, \dot{\Omega}_s \rangle = \langle \xi, (\dot{E}_s - V(v))\Omega_s \rangle.$$

Because $\mathcal{C}$ is an operator core, the definition of the adjoint gives $\dot{\Omega}_s \in D(H_s)$ and

$$H_s\dot{\Omega}_s = (\dot{E}_s - V(v))\Omega_s.$$

Taking the inner product with Ω_s proves the Hellmann–Feynman identity. The parallel gauge and the gap then give the asserted formula for $\dot{\Omega}_s$.

Finally, the preceding norm-continuity argument holds for every $T > 0$. Since E_s is continuous, $e^{-TH_s} = e^{TE_s}e^{-TK(g_s)}$ converges in norm at the endpoints for every $T > 0$. Semigroup convergence gives the explicit resolvent identity

$$(H_s + 1)^{-1} = \int_0^\infty e^{-T}e^{-TH_s} dT.$$

The integrand differences converge in norm pointwise in $T > 0$ and are bounded by $2e^{-T}$. Bochner dominated convergence therefore proves norm-resolvent, hence strong-resolvent, convergence at both endpoints. $\square$

5 The uniform-gap obligation

5.1 What the cited log-Sobolev theorem says

Bauerschmidt–Dagallier [1, Eq. (1.6), Thm. 1.1] use, on a lattice of mesh ε , the Dirichlet form

$$\mathcal{D}_{\text{BD}}(F) = \varepsilon^{-d} \sum_x \mathbb{E} \left[\left(\frac{\partial F}{\partial \varphi_x} \right)^2 \right].$$

It is the unweighted ℓ^2 gradient form of stochastic quantisation. The covariance-weighted form used in `h6-attack.tex` is instead

$$\mathcal{D}_C(F) = \mathbb{E} \langle \nabla F, C \nabla F \rangle, \quad \widehat{C}(k) = \frac{1}{2\sqrt{k^2 + m_0^2}}.$$

Since $0 < C \leq (2m_0)^{-1}I$,

$$\mathcal{D}_C(F) \leq (2m_0)^{-1}\mathcal{D}_{L^2}(F).$$

Thus a covariance-weighted Poincaré inequality implies an unweighted one (with a changed constant), but the converse does not follow: $\widehat{C}(k) \rightarrow 0$ as $|k| \rightarrow \infty$, so no inequality $I \leq cC$ is possible. The mismatch is structural, not a missing constant.

Furthermore, the spectral gap furnished by an LSI is the gap of the Markov generator associated with $\mathcal{D}_{\text{BD}}$. That generator is not the physical Hamiltonian $H(g)$. A separate Euclidean-time correlation argument is required.

5.2 The exact Euclidean-to-Hamiltonian implication

Lemma 5.1 (dense-family decay forces a Hamiltonian gap). *Let $H \geq 0$ have a simple ground state Ω , let $Q = 1 - |\Omega\rangle\langle\Omega|$, and let $\mathcal{D} \subset Q\mathcal{H}$ be dense. Suppose there is $\gamma > 0$ such that for every $\psi \in \mathcal{D}$ there is $C_\psi < \infty$ satisfying*

$$0 \leq \langle \psi, e^{-tH} \psi \rangle \leq C_\psi e^{-\gamma t} \quad (t \geq 0).$$

Then $\text{spec}(H|_{Q\mathcal{H}}) \subseteq [\gamma, \infty)$.

Proof. Fix $0 < \varepsilon < \gamma$ and let $R_\varepsilon = \mathbf{1}_{[0, \gamma-\varepsilon]}(H)Q$. For $\psi \in \mathcal{D}$, the spectral theorem gives

$$\langle \psi, e^{-tH} \psi \rangle \geq e^{-(\gamma-\varepsilon)t} \|R_\varepsilon \psi\|^2.$$

Combining this with the assumed upper bound yields

$$\|R_\varepsilon \psi\|^2 \leq C_\psi e^{-\varepsilon t} \quad (t \geq 0),$$

and therefore $R_\varepsilon \psi = 0$. Density of $\mathcal{D}$ and boundedness of R_ε imply $R_\varepsilon = 0$. Since this holds for every $\varepsilon > 0$, the spectral projection of $H|_{Q\mathcal{H}}$ on $[0, \gamma)$ is zero. $\square$

Decay of the two-point function of the single field φ is not enough to invoke this lemma: it controls only vectors of the form $\varphi(f)\Omega$. Cyclicity of the full local algebra does not turn that one-particle family into a dense family. One needs the same exponent for a collection of centered polynomial/cylinder observables whose vectors are dense.

The unresolved gap statement is consequently:

($\mathbf{G}_{\text{exact}}$). Specify a noncritical class of polynomials/couplings, a cylinder $*$ -algebra $\mathcal{C}$, and $\gamma > 0$ such that, for every $g \in \mathcal{G}$,

$$\overline{\text{span}}\{F\Omega_g : F \in \mathcal{C}\} = \mathcal{H}_0$$

and, for every $F \in \mathcal{C}$, there is $C_{g,F} < \infty$ for which

$$0 \leq \langle Q_g F \Omega_g, e^{-tH(g)} Q_g F \Omega_g \rangle \leq C_{g,F} e^{-\gamma t} \quad (t \geq 0), \quad Q_g = 1 - |\Omega_g\rangle\langle\Omega_g|.$$

Only the exponent γ must be uniform in g and F . Then $\text{span}\{Q_g F \Omega_g : F \in \mathcal{C}\}$ is dense in $Q_g \mathcal{H}_0$, and Lemma 5.1 gives $\text{spec } H(g) \subseteq \{0\} \cup [\gamma, \infty)$ uniformly in g .

The theorem of [1] is uniform in its lattice and volume parameters under its hypotheses, but it treats the unweighted stochastic-quantisation form and constant-coupling finite-volume measures. It does not state ($\mathbf{G}_{\text{exact}}$) for arbitrary profiles $0 \leq g(x) \leq 1$. In a symmetry-broken regime a uniform full-net gap is in any event impossible by the tunnelling argument already proved in `../m3/m3.tex`. Therefore the regime must be part of the gap theorem, not appended after it.

6 The local-second-moment obligation

The remaining moment statement is

$$\sup_{\substack{g \in \mathcal{G}, h \in C_c^\infty(\mathbb{R}; \mathbb{R}) \\ \|h\|_\infty \leq 1, \text{ supp } h \subset I_h, |I_h| \leq 2}} \|V(h)\Omega_g\|^2 < \infty \quad (\text{M}_2)$$

with one constant for all indicated g and h . Proposition 3.1 proves that every term in (M₂) is defined; it does not prove uniform boundedness.

6.1 Why the claimed Poincaré induction is invalid

Let B be a bounded interval, let $C = (2\sqrt{-\partial_x^2 + m_0^2})^{-1}$, and let

$$W_u = M_u C M_u$$

on $L^2(B)$, where $u \in C_c^\infty(B)$ is nonzero. The proof of Theorem 2.3 in `h6-attack.tex` uses both

$$\|v\|_\infty^2 \leq \kappa_B \|v\|_2^2 \quad \text{for all } v \in L^2(B), \quad (6.1)$$

and $\text{Tr } W_u < \infty$. Both assertions are false.

Proposition 6.1 (the two failed estimates). *There is no finite κ_B for which (6.1) holds. Moreover W_u is positive and Hilbert–Schmidt but not trace class; in particular $\text{Tr } W_u = +\infty$.*

Proof. Choose $x_0 \in B$ and a nonzero $\eta \in C_c^\infty((-1, 1))$. For all sufficiently large n ,

$$v_n(x) = n^{1/2} \eta(n(x - x_0))$$

is supported in B , while

$$\|v_n\|_2 = \|\eta\|_2, \quad \|v_n\|_\infty = n^{1/2} \|\eta\|_\infty.$$

This contradicts (6.1).

The kernel $C(x - y)$ has only a logarithmic diagonal singularity, so $u(x)C(x - y)u(y) \in L^2(B \times B)$; hence W_u is Hilbert–Schmidt. To compute its trace, let C_Λ be the positive Fourier multiplier

$$\widehat{C}_\Lambda(k) = \mathbf{1}_{[-\Lambda, \Lambda]}(k) \frac{1}{2\sqrt{k^2 + m_0^2}}.$$

Then $0 \leq M_u C_\Lambda M_u \leq W_u$. More explicitly, $A_\Lambda = C_\Lambda^{1/2} M_u$ is Hilbert–Schmidt because

$$\|A_\Lambda\|_2^2 = \frac{\|u\|_2^2}{2\pi} \int_{-\Lambda}^{\Lambda} \frac{dk}{2\sqrt{k^2 + m_0^2}} < \infty.$$

Hence $M_u C_\Lambda M_u = A_\Lambda^* A_\Lambda$ is positive trace class and

$$\begin{aligned} \text{Tr}(M_u C_\Lambda M_u) &= C_\Lambda(0) \int_B u(x)^2 dx \\ &= \frac{\|u\|_2^2}{2\pi} \int_{-\Lambda}^{\Lambda} \frac{dk}{2\sqrt{k^2 + m_0^2}}. \end{aligned}$$

The last integral diverges logarithmically as $\Lambda \rightarrow \infty$. If W_u were trace class, monotonicity of the trace on positive operators would bound all these traces by $\text{Tr } W_u$, a contradiction. $\square$

The invalid step in `h6-attack.tex` is precisely

$$\sum_{\alpha} \lambda_{\alpha} \|e_{\alpha}\|_{\infty}^2 \lesssim \sum_{\alpha} \lambda_{\alpha} = \text{Tr } W_u.$$

Proposition 6.1 shows that neither inequality is available. In addition, the spectral vectors e_{α} are a priori L^2 vectors, whereas the induction was stated only for smooth test functions; an approximation/closability argument was not supplied. The claimed implication “(P-U) $\Rightarrow$ (H6)” is therefore withdrawn.

6.2 A corrected conditional recursion

The mean terms can be summed without $\text{Tr } W_u$, but doing so exposes an additional input. The following proposition records the valid statement.

Proposition 6.2 (corrected covariance recursion). *Fix a bounded interval B and an integer $N \geq 1$. Suppose, uniformly in g , that:*

1. *the covariance-weighted Poincaré inequality*

$$\text{Var}_{\mu_g}(F) \leq c_P \mathbb{E}_{\mu_g} \langle \nabla F, C \nabla F \rangle$$

holds for the ultraviolet-mollified Wick polynomials introduced below;

2. *for $0 \leq n \leq N$ the mollified one-point functions satisfy*

$$m_{n,g,\varepsilon}(x) = \mathbb{E}_{\mu_g}[\varphi_{\varepsilon}(x)^n], \quad \sup_{\substack{g \in \mathcal{G} \\ 0 < \varepsilon \leq 1}} \|m_{n,g,\varepsilon}\|_{L^{\infty}(B)} \leq b_n;$$

3. *for every $u \in C_c^{\infty}(B)$ and $n \leq N$, $F_{n,\varepsilon}[u] \rightarrow F_n[u]$ in μ_g -probability as $\varepsilon \downarrow 0$, where*

$$F_{n,\varepsilon}[u] = \int_B u(x) : \varphi_{\varepsilon}(x)^n : dx, \quad F_n[u] = \int_B u(x) : \varphi(x)^n : dx.$$

Then, for every $n \leq N$,

$$\sup_{\substack{g \in \mathcal{G}, \\ u \in C_c^{\infty}(B; \mathbb{R}) \\ \|u\|_{\infty} \leq 1}} \mathbb{E}_{\mu_g}[F_n[u]^2] < \infty.$$

Consequently (M_2) holds for every polynomial of degree at most N ; translation covariance makes the constant independent of the location of the interval B .

Proof. Choose a nonnegative, compactly supported smooth approximate identity ρ_{ε} and put $\varphi_{\varepsilon} = \rho_{\varepsilon} * \varphi$ and

$$C_{\varepsilon} = \rho_{\varepsilon} * C * \tilde{\rho}_{\varepsilon}, \quad \tilde{\rho}_{\varepsilon}(x) = \rho_{\varepsilon}(-x).$$

Wick ordering at scale ε is with respect to $C_{\varepsilon}(0)$. Let $K_{n,g,\varepsilon}$ denote the covariance kernel of the mollified n th Wick power, so

$$\text{Var}_{\mu_g}(F_{n,\varepsilon}[u]) = \langle u, K_{n,g,\varepsilon} u \rangle.$$

All inequalities below are inequalities of positive kernels, equivalently of their quadratic forms on test functions. Set

$$S_{1,g,\varepsilon} = c_P C_{\varepsilon}.$$

For $n \geq 2$ define recursively the positive kernel

$$S_{n,g,\varepsilon}(x, y) = c_P n^2 C_{\varepsilon}(x - y) (S_{n-1,g,\varepsilon}(x, y) + m_{n-1,g,\varepsilon}(x) m_{n-1,g,\varepsilon}(y)). \quad (6.2)$$

We prove $0 \leq K_{n,g,\varepsilon} \leq S_{n,g,\varepsilon}$ by induction.

For $n = 1$, the covariance-weighted norm of the gradient is $\langle u, C_\varepsilon u \rangle$, so Poincaré gives $\langle u, K_{1,g,\varepsilon} u \rangle \leq c_P \langle u, C_\varepsilon u \rangle$. Assume $K_{n-1,g,\varepsilon} \leq S_{n-1,g,\varepsilon}$. The chain rule gives

$$\frac{\delta F_{n,\varepsilon}[u]}{\delta \varphi(z)} = n \int_B u(x) \rho_\varepsilon(x-z) : \varphi_\varepsilon(x)^{n-1} : dx.$$

Degree lowering and Poincaré therefore give

$$\begin{aligned} \langle u, K_{n,g,\varepsilon} u \rangle &\leq c_P n^2 \iint u(x) u(y) C_\varepsilon(x-y) \\ &\quad \times (K_{n-1,g,\varepsilon}(x,y) + m_{n-1,g,\varepsilon}(x) m_{n-1,g,\varepsilon}(y)) dx dy. \end{aligned}$$

The Schur product theorem applies to positive kernels: multiplying the positive-form inequality $K_{n-1,g,\varepsilon} \leq S_{n-1,g,\varepsilon}$ by the positive-definite kernel C_ε preserves the inequality. The right side is therefore at most $\langle u, S_{n,g,\varepsilon} u \rangle$, proving the induction.

The ground-state derivative formula is not involved. Iterating (6.2) instead shows that $S_{n,g,\varepsilon}$ is a finite sum of terms, each bounded in absolute value on $B \times B$ by

$$A \sum_{r=1}^n C_\varepsilon(x-y)^r,$$

where A depends only on $n, c_P, b_0, \dots, b_{n-1}$. Every finite power of the logarithmically singular kernel $C(x-y)$ is locally integrable in one dimension, and the bound is uniform in ε . Indeed, with $\kappa_\varepsilon = \rho_\varepsilon * \tilde{\rho}_\varepsilon$ a probability density, $C_\varepsilon = \kappa_\varepsilon * C$; Jensen's inequality gives, for every integer $r \geq 1$,

$$C_\varepsilon(z)^r \leq (\kappa_\varepsilon * C^r)(z).$$

After enlarging the fixed compact difference set $B - B$ by the support of κ_ε , integration shows

$$\sup_{0 < \varepsilon \leq 1} \int_{B-B} C_\varepsilon(z)^r dz < \infty.$$

Hence $\langle |u|, S_{n,g,\varepsilon} |u| \rangle$ is bounded uniformly in g and ε . The mean obeys

$$|\mathbb{E} F_{n,\varepsilon}[u]| \leq b_n \|u\|_{L^1(B)},$$

so $\sup_{g,\varepsilon} \mathbb{E}[F_{n,\varepsilon}[u]^2] < \infty$.

For fixed g , convergence in probability supplies a sequence $\varepsilon_j \downarrow 0$ along which $F_{n,\varepsilon_j}[u] \rightarrow F_n[u]$ almost surely. Fatou's lemma yields

$$\mathbb{E}_{\mu_g}[F_n[u]^2] \leq \liminf_{j \rightarrow \infty} \mathbb{E}_{\mu_g}[F_{n,\varepsilon_j}[u]^2],$$

with the same bound independent of g . A finite sum and Cauchy–Schwarz handle a polynomial P . $\square$

The extra uniform L^∞ hypothesis on the mollified one-point functions in Proposition 6.2 is not supplied by the first-moment estimate quoted in `h6-attack.tex`. A bound $|\mathbb{E} F_n[u]| \leq c \|u\|_\infty$ identifies a finite signed measure, not an L^∞ density; an atomic measure has infinite logarithmic self-energy and would not close (6.2). This is the exact distinction hidden by the false inequality (6.1).

There are therefore two honest routes to (M_2) :

1. prove a local N_τ /hypercontractive estimate directly, uniform in the remote spatial cutoff;
or

2. prove both the covariance-weighted Poincaré inequality and the uniform one-point density bounds in Proposition 6.2.

The unweighted LSI of [1] supplies neither item 1 nor the covariance-weighted inequality in item 2.

7 Corrected proof program

The dependency order is now:

$$\boxed{\text{classical cutoff construction}} \implies \boxed{\text{finite propagation} + \Omega_g \in D(V(h))}$$

and, independently,

$$\begin{aligned} & \boxed{(G_{\text{exact}})} \implies \boxed{\text{FKN path regularity, Theorem 4.6}}, \\ & \boxed{(G_{\text{exact}})} + \boxed{(M_2)} \\ \implies & \boxed{\text{LPPL cell estimate in lpp1.tex}} \implies \boxed{\text{full-net local-norm convergence}}. \end{aligned}$$

The last implication includes local normality, translation invariance, $\mathbb{Z}_2$ invariance for even P , and the algebraic ground-state property, exactly as proved in `lpp1.tex` and `../m0/m0.tex`.

The remaining work is to be carried out in the following explicit order.

1. **Path theorem (executed).** Theorem 4.6 proves norm- C^1 dependence of $e^{-TK(g_s)}$ directly from the Feynman–Kac–Nelson formula. Under (G_{exact}) , spectral perturbation then gives norm-continuous ground-state projections, the Hellmann–Feynman identity, the ground-state-vector derivative, and endpoint strong-resolvent continuity. No common form domain is needed.
2. **Regime-specific physical gap.** State the coupling/mass region before proving a gap. Establish uniform Euclidean-time connected decay, with one exponent, for a dense cylinder algebra and for every profile $0 \leq g \leq 1$. Apply Lemma 5.1. A stochastic-quantisation spectral gap alone is not a substitute.
3. **Local second moment.** Prove (M_2) by a uniform local higher-order estimate, or verify every hypothesis of Proposition 6.2. In the latter route, the required Poincaré form is covariance-weighted and the one-point functions must have uniformly bounded densities; a total-variation bound is insufficient.
4. **Assembly (executed).** The FKN path theorem, Proposition 2.1, and Proposition 3.1 have been inserted into `lpp1.tex`. Its cell estimate now assumes only (G_{exact}) and (M_2) . The derivative is integrated on $[\varepsilon, 1 - \varepsilon]$ and the endpoints are obtained by norm continuity of the ground-state projection.
5. **Wavelet uniformity only afterward.** Once the continuum theorem is unconditional in a stated regime, prove discrete propagation, gap, and moment estimates with constants uniform in resolution, and derive path regularity from the corresponding discrete FKN formula. The continuum gap and the stochastic-quantisation gap must not be identified.

This ordering removes completed work from the hypothesis list, prevents the endpoint-domain problem from contaminating the interior derivative, and makes the two genuinely new estimates — physical gap and local second moment — visible as separate theorems.

8 Final status theorem

Theorem 8.1 (rigorous status of M4). *Within the standard spatial-cutoff $P(\varphi)_2$ construction:*

1. *the actual cutoff dynamics satisfies exact finite propagation with speed at most one;*
2. *every cutoff ground state belongs to the domain of every compactly smeared Wick polynomial;*
3. *(G_{exact}) implies the affine-path regularity and endpoint continuity needed in the state-derivative argument;*
4. *(G_{exact}) and (M_2) imply all conclusions of the conditional continuum LPPL theorem in `lppl.tex`;*
5. *neither the uniform physical gap nor (M_2) follows from the Bauerschmidt–Dagallier LSI;*
6. *the proof of “ $(P-U) \Rightarrow (H6)$ ” in `h6-attack.tex` is invalid for the reasons in Proposition 6.1.*

Consequently the main spatial-cutoff convergence theorem is not yet unconditional. Its remaining inputs are exactly (G_{exact}) and (M_2) .

Proof. Items 1 and 2 are Propositions 2.1 and 3.1. Item 3 is Theorem 4.6. For item 4, that theorem supplies the derivative formula and endpoint continuity, Proposition 2.1 supplies the exact light cone, (G_{exact}) supplies the spectral filter and clustering estimate, and (M_2) supplies the uniform cell norm. These are exactly the inputs used in the proof of the one-path estimate in `lppl.tex`; no hypothesis (D) or (L) remains. Items 5 and 6 were proved in §5 and Proposition 6.1. The last sentence is the resulting exhaustive list. $\square$

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